

Statistics

Histogram: It is a graphical representation of the distribution of numerical data.

It is an estimate of probability distribution of a continuous variable.

It is used to visualize the shape, central tendency & variability of the dataset.

Ages = { 11, 12, 14, 18, 24, 26, 30, 35, 36, 37, 40, 41, 42, 43, 50 } → histogram

11-50

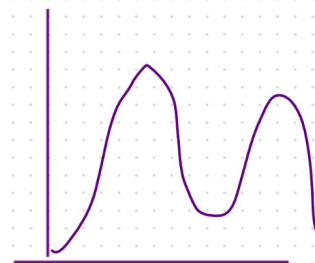
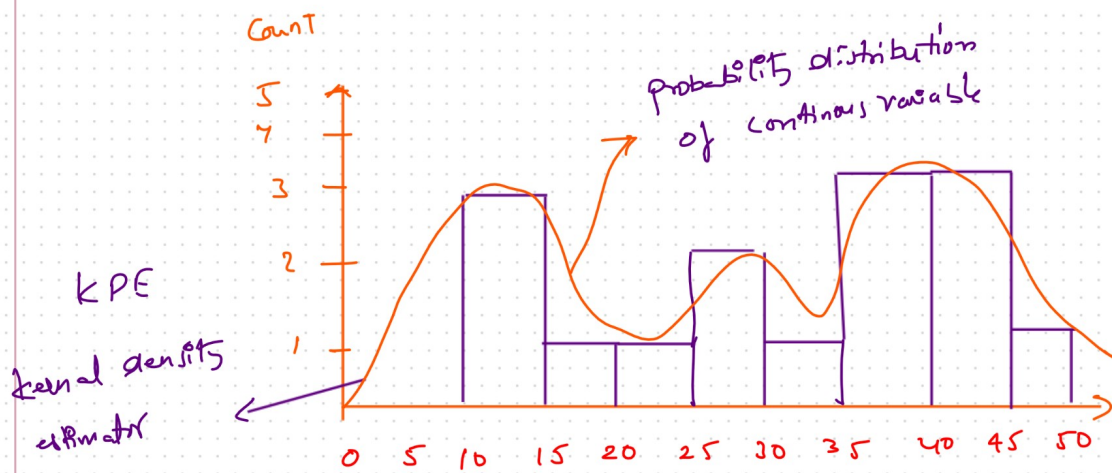
∴ No. of bins = 10

$$\frac{50}{10} = 5 \text{ (bin size)}$$

$$(5) = \frac{50}{5} = 10$$

Bins [0-5, 5-10, 10-15, 15-20, ..., 45-50]

Frequency	0-5 = 0	20-25 = 1	40-45 = 3
	5-10 = 0	25-30 = 2	45-50 = 1
	10-15 = 3	30-35 = 1	
	15-20 = 1	35-40 = 3	



Histogram

symmetric data

asymmetric data



Skewness : It measures the asymmetry of a data distribution

It answer is the data balanced (symmetric) or tilted to one side?

cleaned data set

→ mean tells us where the center is

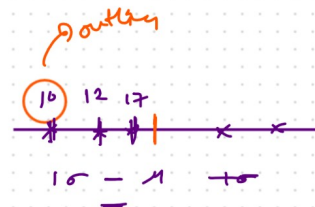
→ S.d tell us how spread out the data's

→ Neither tells you whether the data is

symmetric or asymmetric (lopsided)

∴ Skewness fills the gap — by showing the

Shape & direction of The distribution



$$\mu = 20$$

6-5

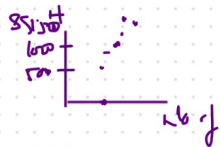
$$10 \cdot 2 \cdot 4 - 6 =$$

$$20 - 5 = 15$$

$$\mu = 2\sigma \approx 10$$

$$u - 35 = 5$$

5 ~ 10



Case 1: (No-skew)

House prediction

x_2	y_3
<u>Sp. fed</u>	locations
	Area

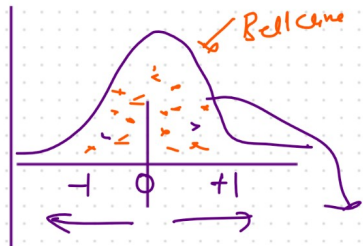
600g —
0.0125

Y
Cost

251

$$0.12500$$

652 (78)
 159 (11)
 90 (80%)
 9 (100%)

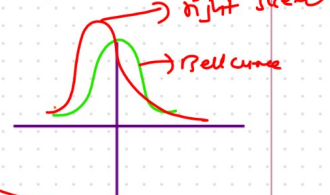


0.245

0. P45

0: 947

Normal distribution



Case 2

Right Skew (Positive Skew)

$$X = \{10, 20, \overset{\downarrow}{30}, 40, 90\}$$

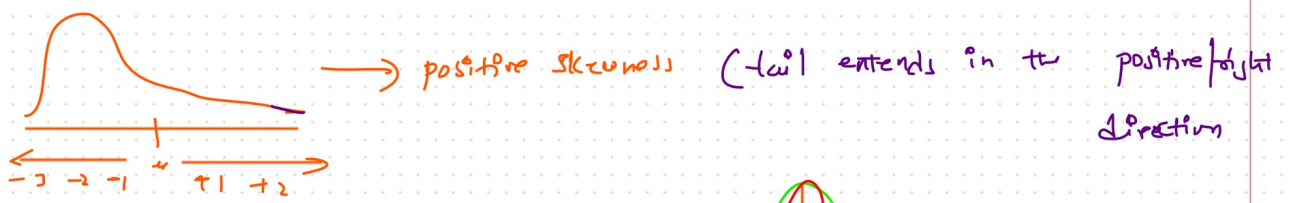
Mean = 38

Median = 30

mean > median

→ The high outlier (90) pulls the mean towards the right

Why Mean $>$ median $\rightarrow$ cause mean is pulled by the extreme value (90)



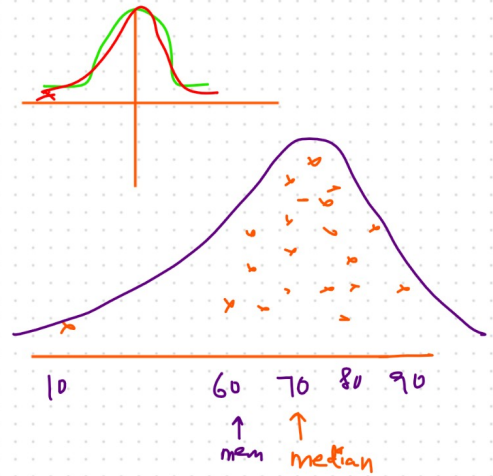
Case 3: Left skew (Negative skew)

$$x = \{10, 60, 70, 80, 90\}$$

$$\text{Mean} = 62 \quad \text{Median} = 70$$

$$\rightarrow \text{mean} < \text{median}$$

$\rightarrow$ The lower outlier (10) pulls the mean to the left



Skewness Formula (Fisher's Moment Coefficient)

$$g_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\underbrace{\quad}_3}$$

$$\text{ages } \{10, 20, 30\}$$

$$(-1)^3 + (+1)^3 = +ve$$

$$(-1)^3 = -1$$

$$g_1 = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\underbrace{\quad}_3} \right)^3$$

$$g_1 > 0 \rightarrow \text{Right skew}$$

$$g_1 < 0 \rightarrow \text{Left skew}$$

$$g_1 \approx 0 \rightarrow \text{Symmetric / Balanced}$$

$$\begin{matrix} 0.245 \\ 0.154 \\ \textcircled{0.75} \end{matrix}$$

Kurtosis : It measures how much extreme values (outliers) are present in a dataset, It measures the heaviness of the tails of a distribution

$\therefore$ extreme values will appear more or less frequently than in normal distribution

$\rightarrow$ Are outliers common, rare, normal

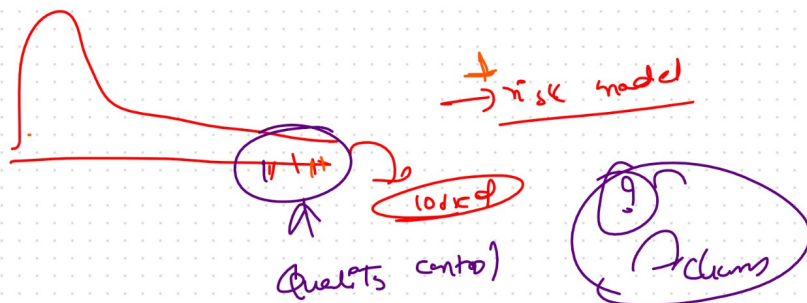
Skewness

1, The data tilted? which direction? tail

Kurtosis

are extreme values (outliers) common in this data

scipy.stats.kurtosis()



Excess kurtosis = Raw kurtosis - 3

normal
 -3.0
 -2.0
 -1.0

Mesokurtic

Normal-like balanced distribution

Leptokurtic

extreme values are more common [more outliers]

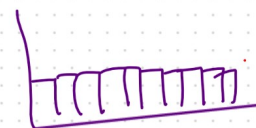
Platykurtic

extreme values are less common than normal

Formula

$$g_2 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4}{s^4} - 3$$

{ 16, 20, 30, 40, 50, 60, 70, 80 }



$$g_2 = \frac{1}{n} \left(\frac{\sum_{i=1}^n (x_i - \bar{x})^4}{s^4} \right) - 3$$

$x = \{ \dots \}$

Age = { 22, 20, 22, 24, 20, 22 }

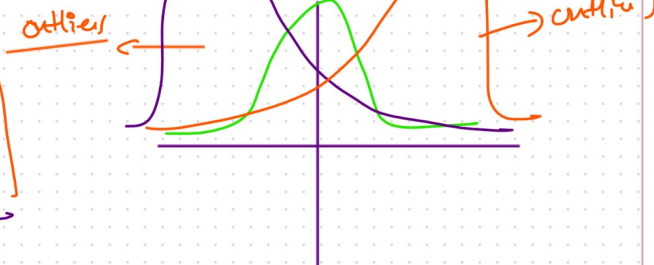
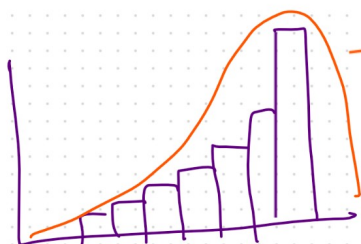
mu = avg

KDF

frequency



mean > median
 mean < median



HP

Not any

z. test

Any

predict

5, 1, 10, 20, 22, 24, 26, 28, 30

How many outliers

2

1000

Negative

2

1500

Kelvin

3

200



Mathematical

-2

-3

-4

$(-1)^2 = 1$ $(-1)(-1)^2 = -1$

2 (variance)

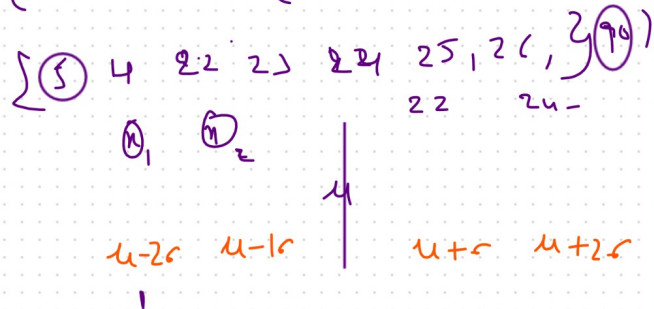
always true sign

3 (skewness)

preserve sign

4 (kurtosis)

always true sign



Deviation

Squared (power of 2)

Cubed (power of 3)

4th power

0.2
5

0.04

25

0.008

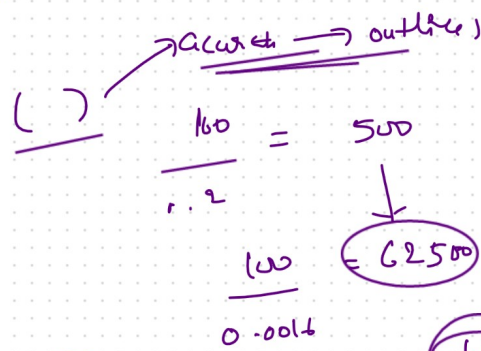
125

0.0016

625

5 is only 25 larger than 0.2

625 is 100's of larger than 0.2



S.D. (1, 2, 3) = 2 units

outlier (1, 4, 9) = 8 units

1 8 27 = 26 units

(2) = 75%

(3) = 40%

(4) = 100%

Higher power = larger separation

= better differentiator of outlier

1.5

99.3%

$Q_3 - Q_1$

$$\Sigma Q_R = Q_3 - Q_1$$

0.7%

